

SHRESHTH VERMA

Full-Stack & AI Engineer

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SUMMARY

I started programming when I was 12 and have been building software ever since. During college I founded a software agency that delivered products for startups across India and international clients. Over the past few years I have focused on AI systems browser automation, scalable backend infrastructure and developer tooling.

EXPERIENCE

SDE Intern | Vingo Aug 2026 – Sep 2026

Peer-to-peer marketplace for buying, selling and bidding on items. TypeScript, NestJS, PostgreSQL, Prisma, Redis, GCP Cloud Run, Next.js, React, TanStack Query

- Architected the **server-driven UI (SDUI)** layer behind the mobile app's bidding and product screens, so UI and business rules ship from the backend with zero app-store releases. Built on a hexagonal (ports & adapters), CQRS-style NestJS backend.
- Found and fixed a scaling flaw in API rate limiting: limits were applied per instance, so effective capacity **grew N× under Cloud Run autoscaling**. Made them a single distributed limit backed by Redis, with fail-closed error handling.
- Designed the marketplace's core product feed with keyset (cursor) pagination, recursive CTE category-tree traversal and multi-dimensional filtering in PostgreSQL, keeping **pagination stable and deterministic** across sort orders.
- Owned the **internal admin platform** end to end: orders, payments, shipments, user management, content moderation and a 6-step listing workflow with Google Maps geolocation. Stack: Next.js, TanStack Table/Query, React Hook Form + Zod.
- Led the public website's redesign and technical SEO strategy (dynamic sitemaps, Open Graph, crawler controls), and built fuzzy search ranked by subsequence distance.

Co-Founder | Olange Apr 2026 – Jul 2026

Built an AI-native operating system for real estate teams — unifying WhatsApp CRM, lead management, listings, site-visit scheduling, and AI workflows into one platform.

- Architected the platform end to end on modern cloud infrastructure with a scalable, multi-tenant backend and real-time collaboration.
- Built AI-powered features including lead qualification, WhatsApp automation, document intelligence, and property image enhancement (AI staging) to streamline sales operations.
- Lead product strategy, engineering, customer discovery, and go-to-market, working directly with real estate developers and brokerages to shape the roadmap.

Co-Founder | DB Stencil Nov 2025 – Mar 2026

- Founded and built a visual database modeling platform, growing to **500+ users** with real-time collaborative schema design and instant code generation.
- Architected a schema engine supporting **50+ database operations** — tables, relationships, indexes, constraints, and advanced modeling features.
- Built multi-framework code generation for DBML, Prisma, Drizzle ORM, and SQL DDL, accelerating database development workflows.
- Designed a robust state-synchronization and persistence system for complex diagrams, enabling seamless editing, versioning, and project management.

AI Automation Engineer | YesMadam Jul 2025 – Aug 2025

- Built a high-throughput background processing system handling **600K+ image verification requests per month** for employee compliance workflows.
- Designed resilient asynchronous pipelines using Inngest with automatic retries, rate limiting, and queue orchestration.
- Optimized processing reliability and throughput while reducing operational overhead; containerized services with Docker for consistent, scalable deployments.

Freelance - Backend Developer | Pnacademy Jun 2024 – Dec 2024

- Built and deployed backend services serving **1,000+ monthly active users** using Node.js, TypeScript, PostgreSQL, and Redis.
- Reduced search latency from 800ms to 200ms (**75%**) using PostgreSQL full-text search (tsvector); improved API response times **40–60%** with Redis caching.
- Implemented JWT authentication and RBAC (3+ roles) and AWS S3 uploads (500+ files/day) via pre-signed URLs; deployed Dockerized services on AWS EC2 (Nginx, SSL) at **99%+ uptime**.

SELECTED PROJECTS

Autonomous Browser Agent [\[Demo\]](#) [\[Github\]](#)

- General-purpose AI agent that executes plain-English tasks by driving a real anti-detect Firefox browser (Camoufox + Playwright) end-to-end — shopping, research, form-filling, and checkout — with a narrating REPL and human confirmation before consequential actions.

- Model-agnostic perceive–act loop (any LLM via OpenRouter) grounded in the page's computed accessibility tree (aria_snapshot) plus per-turn vision, **resolving custom click-targets, iframes, and shadow DOM**; hardened with a supervisor-LLM anti-loop, injection-proof action guards, CAPTCHA handoff, and history compaction + prefix caching for near-flat token cost. Built in Python.

Timbre — Voice AI Platform [\[Live\]](#) [\[Github\]](#)

- Real-time voice AI platform to deploy conversational agents on any website with a single script tag.
- Achieved **sub-500ms voice-to-voice latency** using WebRTC, streaming STT (Deepgram), LLM (OpenRouter), and TTS (Cartesia); RAG via pgvector, MCP tool integrations, and HTTP function calling. Backend on Hono (Bun), Supabase Postgres, and LiveKit.

EDUCATION & ACHIEVEMENTS

B.E. Computer Science, Chitkara University 2022 – 2026

- Winner — Chitkara Sandbox 2.0 Hackathon (Mar 2024) · Winner — ShopClues Smart Cataloging Hackathon (Mar 2024).